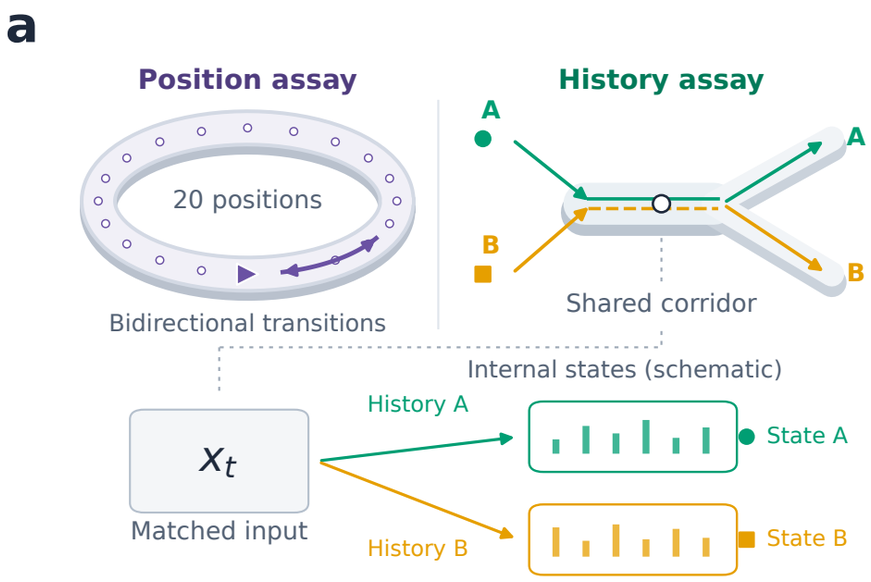
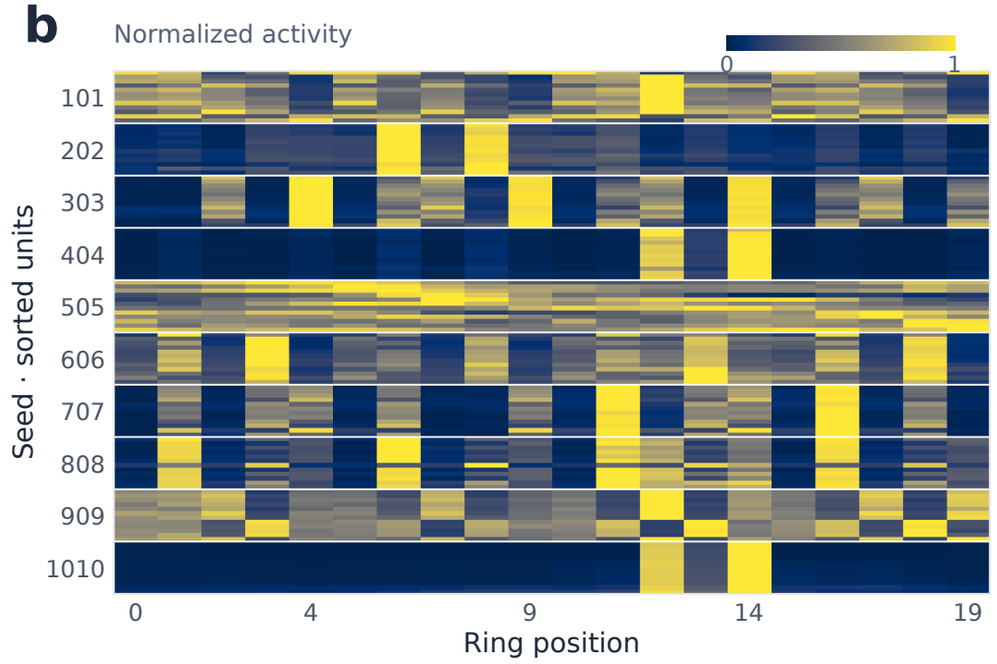
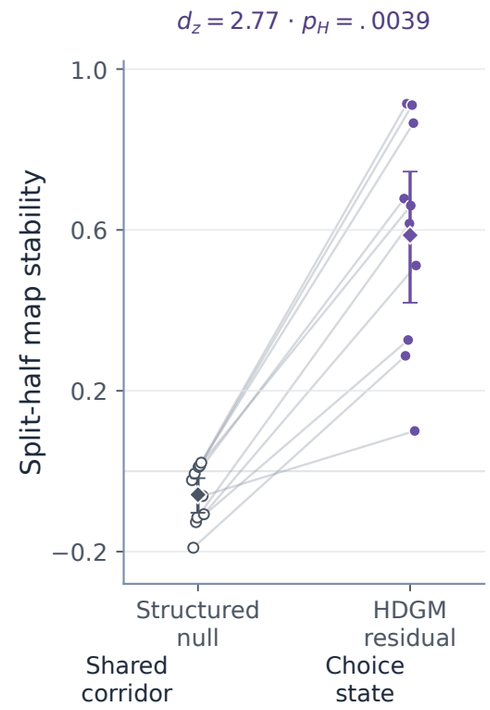
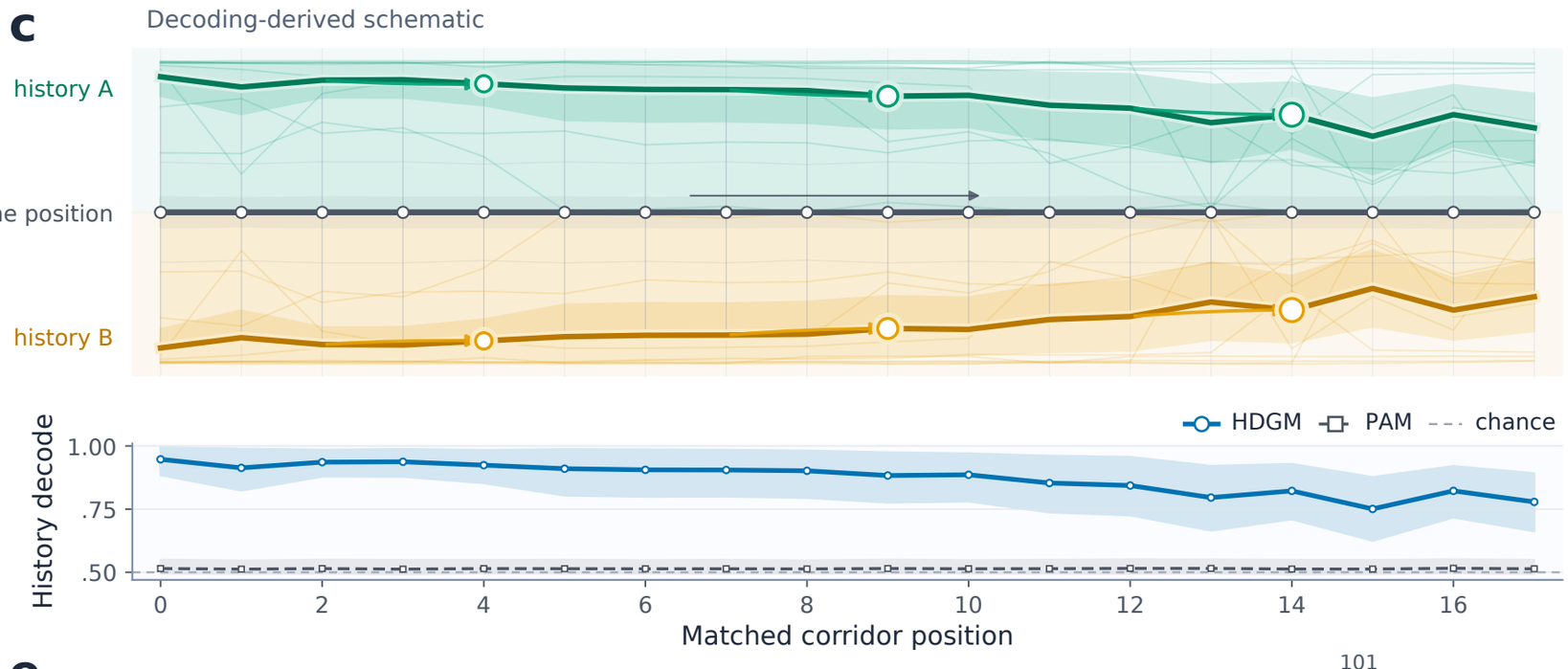
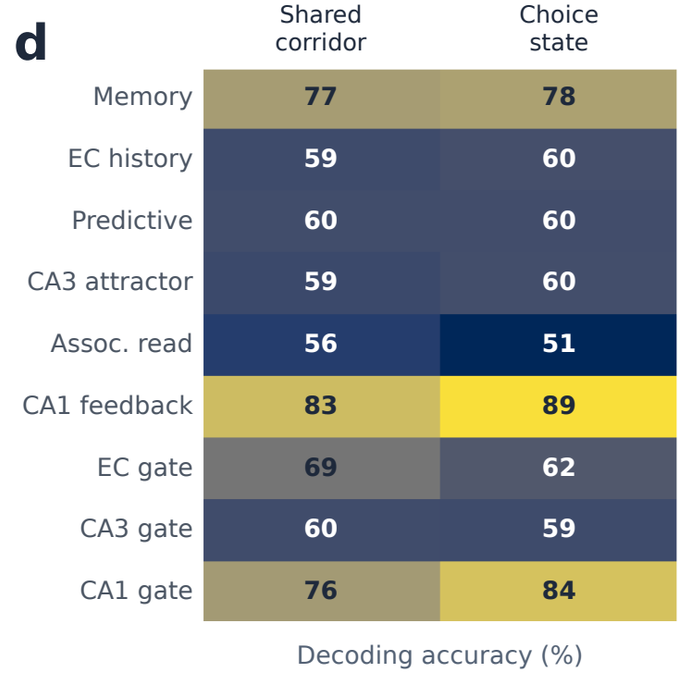
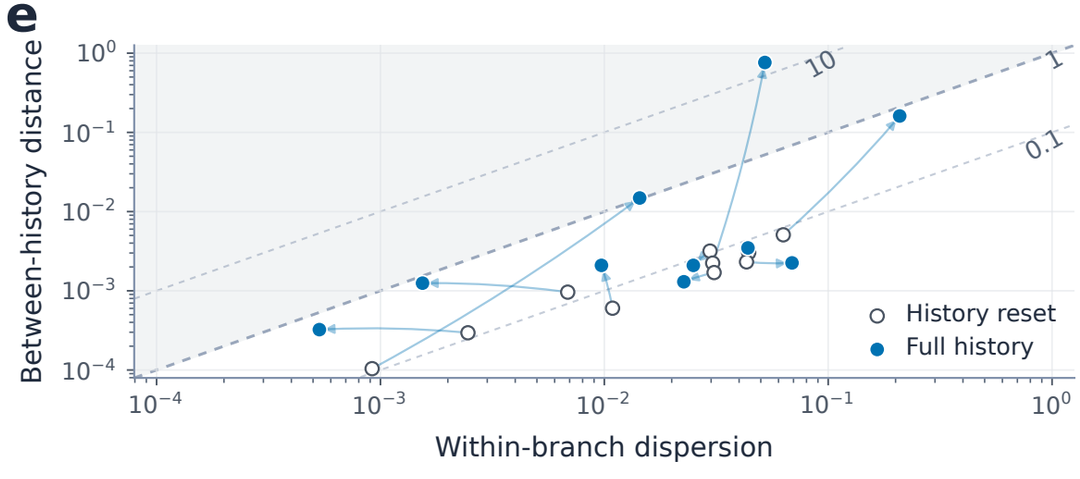

a
Position assay
20 positions
Bidirectional transitions
History assay
A
B
A
B
Shared corridor
Internal states (schematic)
x_t
Matched input
History A
History B
State A
State B

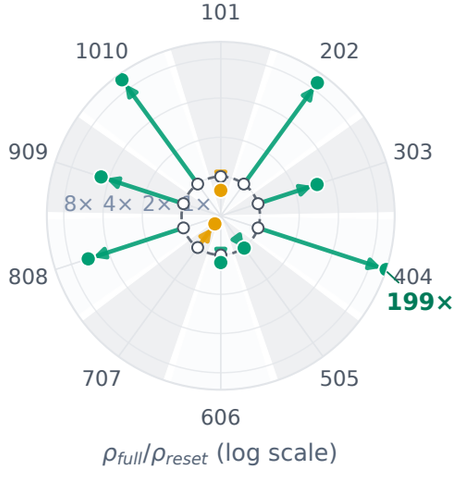

b
Normalized activity
0
1
Seed · sorted units
101
202
303
404
505
606
707
808
909
1010
0
4
9
14
19
Ring position

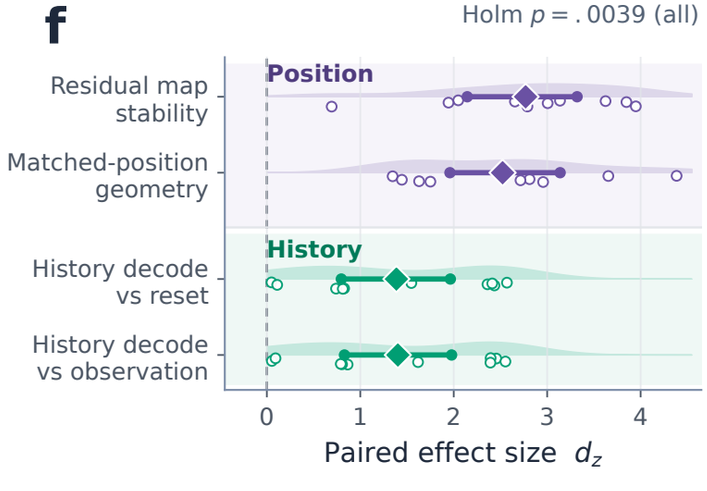

$d_z = 2.77 \cdot p_H = .0039$
Split-half map stability
1.0
0.6
0.2
−0.2
Structured null
HDGM residual
Shared corridor
Choice state


c
Decoding-derived schematic
history A
same position
history B
HDGM
PAM
chance
History decode
1.00
.75
.50
0
2
4
6
8
10
12
14
16
Matched corridor position


d
Memory 77 78
EC history 59 60
Predictive 60 60
CA3 attractor 59 60
Assoc. read 56 51
CA1 feedback 83 89
EC gate 69 62
CA3 gate 60 59
CA1 gate 76 84
Decoding accuracy (%)


e
Between-history distance
Within-branch dispersion
History reset
Full history
10
1
0.1


101
202
303
404
505
606
707
808
909
1010
8× 4× 2× 1×
199×
ρ_{full}/ρ_{reset} (log scale)


f
Holm $p = .0039$ (all)
Residual map stability
Matched-position geometry
History decode vs reset
History decode vs observation
Position
History
0
1
2
3
4
Paired effect size d_z